

Cristobal Cortés Gómez

Game Developer & Mathematician

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Professional Profile

Cross-platform game developer across desktop, mobile and web, shipping indie titles since 2012, with a focus on game feel and optimization for low-end devices. Theoretical Mathematics graduate with deep knowledge of linear algebra, optimization and AI; advanced English, used daily with U.S.-based clients.

Technical Skills

Programming languages: TypeScript/JavaScript, Python, C, C#, GML, GLSL.

Web, frameworks & tools: HTML/CSS, Node.js, React, Redux, Three.js, Firebase.

Game Engines: GameMaker, Unity.

Systems & hardware: Linux (shell scripting), Git; ESP8266, Arduino.

Spoken languages: Spanish (native); English (advanced).

Experience

Jun. 2025–Jun. 2026: Game Developer — *Gnarled Helix LLC* (~10-person U.S. indie studio), remote contract

- Gameplay, AI, shaders and UI for a browser-based 2D action game that teaches programming through an in-game code editor, built on the studio's custom engine (TypeScript, Three.js, React, Redux, WebAssembly physics).
- Authored multiple enemy AIs on the engine's behaviour-tree framework, built the inventory UI in React, wrote a custom god-ray shader for a cinematic level, and added a short grace timer to the water jump to improve swimming.
- Reworked the player's mouse-driven parabolic aiming, deriving the launch solution by hand to cover cases where the quadratic has no real root.

2024–2025: Sole Developer — *Jalapeño Labs*, contract

- Developed the 3D digital adaptation of the board game “Camino a Xibalbá PITZ”, a chess-like strategy game inspired by the Mesoamerican ballgame (GameMaker, JavaScript, Firebase).
- Built the rule-resolution system and full turn-based online multiplayer with user accounts and match records, backed by Firebase Realtime Database.

Additional Experience

2022–2024: Automation, embedded and tutoring contracts, alongside completing the Mathematics degree.

2020–2022: Hardware & Software Developer — *Proyecto INTERSEX*

Built an LED face mask and a 3D museum simulator for a social and artistic project (JavaScript, Arduino, ESP8266, Unity).

Selected Independent Projects

The games below were made with the 4-member Axolotl Bros team; all listed on my portfolio.

2019 (paused) “Libélula” Multiplayer fighting game. Custom 2D light/shadow system, parallax backgrounds, and reactive competitive gameplay tuned to details like coyote time and variable-height jumps.

2018 “Sacrifice Your Brothers” (programmer, 2-person code team) Local multiplayer zombie survival game. Optimized A* pathfinding and game feel work: frame-skipping screen shake, tuned friction/damping, and per-weapon recoil. 4k downloads on itch.io.

2014–2018 (unreleased) “Gravity Box” Multiplayer experimental arcade game. Multi-gravity movement built on the Extreme Physics library, platformer AI with parabolic shots, and a large gameplay-modifier item set.

2014 “Sweet Valentine’s Day” (sole programmer) Arcade parody about love. Mobile adaptability, a Lemmings-like single-pixel collision system, and online highscores; optimized to run on low-end Android hardware.

2012 “Morbus” (sole programmer) Zombie survival game. Simple lighting system, multi-resolution (hierarchical) A* pathfinding for performance, and a 2D top-down bone animation system.

Global Game Jam (48-hour jams): 2012, 2013, 2014, 2016, 2018, 2019, 2020, 2024; local-site honorable mention in 2019.

Education

2025: Bachelor’s Degree in Mathematics — Universidad de Guadalajara (CUCEI). Final project: “Suitability Analysis of Shallow vs. Deep Neural Networks Using the Perceptron for Classification Problems”, built on a self-made neural network in C and CUDA.